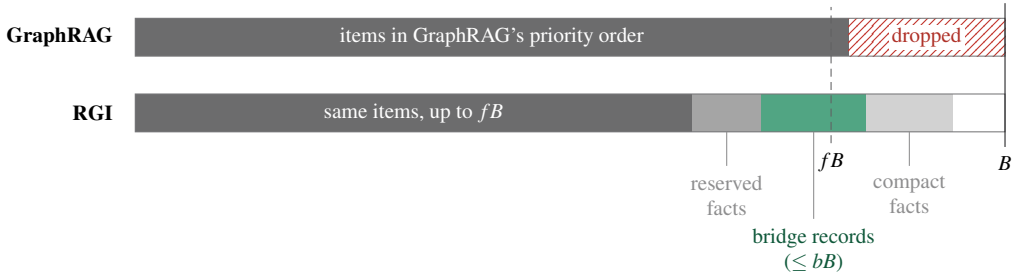

Residual store: every relationship still not in an input, keyed by the communities it touches.
Coverage certificate: represented $\cup$ bridged $\cup$ stored $= E$, with the three sets disjoint, or the build fails.